

A/B Testing Plan: AI-Powered Customer Service Chatbot

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Document Owner: Emily Schmidt

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1. Business Problem

Customers who engage with the call center experience lower satisfaction and higher churn rates relative to industry benchmarks, indicating that the current service experience is a meaningful driver of churn. Call center capacity also limits visibility into the long tail of customer issues, making root-cause analysis slow and incomplete.

We are evaluating whether an **AI-powered chatbot** — embedded in the mobile app and capable of resolving common issues autonomously while escalating to human agents when needed — can reduce churn, improve satisfaction, and surface structured insight into customer pain points. This experiment is a pre-market validation of the product concept, not a launch of a fully scaled solution.

2. Hypothesis

If we introduce an AI-powered chatbot as a support channel for mobile users, then:

- Resolution speed will improve for a meaningful share of contacts (containment without human handoff),
- CSAT will increase among supported users, and
- Churn will decrease in the treatment population,

while producing structured conversation data that improves visibility into the root causes of customer issues.

3. Experiment Design

We will build a **lightweight (MVP) version of the AI chatbot** sufficient to test the core hypothesis. The goal is not to validate a production-grade product but to determine whether AI-assisted support drives meaningful business impact before further investment.

- **Randomization unit:** User
- **Assignment:** Random 50/50 split, stratified by tenure and prior support contact volume to balance pre-existing churn risk
- **Control:** Existing call center support only
- **Treatment:** Existing call center support **plus** access to the AI chatbot in the mobile app, with seamless escalation to a human agent when the bot cannot resolve the issue
- **Duration:** Minimum 6 weeks (covers a full billing/support cycle); pre-registered with a power analysis sized to detect the MDE on churn
- **Pre-registration:** Hypothesis, primary metric, MDE, and decision rules are locked before launch

The design measures both **adoption** and **impact** without forcing channel changes. Because not all users will need support during the experiment, analysis includes both **intent-to-treat (ITT)** and a **triggered analysis** for users who actually engaged with support.

4. Eligibility

- Active mobile app users (logged in within the last 30 days at assignment)
- Located in regions where the chatbot's supported languages and regulatory posture are valid
- Excludes: enterprise/B2B accounts, employees, users in other concurrent experiments touching support flows

This ensures both arms have equivalent ability to access the chatbot, reducing bias from call-center-only users.

5. Unit of Analysis

- **Primary (ITT):** All eligible users assigned to control vs. treatment, regardless of whether they used support — measures overall business impact.
- **Secondary (Triggered):** Users who initiated any support contact during the experiment window — measures the chatbot's effectiveness among the relevant population.

ITT is the decision-driving analysis; triggered analysis is interpretive.

6. Treatment Exposure

A treatment user is considered **exposed** when the chatbot entry point is rendered in their app session. Deeper engagement is tracked separately:

- Chatbot initiation rate
- Conversation completion rate
- Containment rate (resolved by AI without human handoff)
- Escalation rate to human agent

7. Metrics

Primary

- **Churn rate**, treatment vs. control, ITT across all eligible users
- Secondary lens: churn among users who engaged with any support channel

Secondary

- **Chatbot adoption rate:** % of treatment users initiating chat when seeking support
- **Containment rate:** % of chatbot conversations resolved without human escalation
- **CSAT** for support interactions, by channel and overall
- **Time to resolution**, by channel
- **Channel mix shift:** chat vs. call distribution within the treatment arm

Guardrails (must not regress materially)

- Mobile app crash and error rate
- Call center performance (dropped calls, wait times, abandonment)
- Customer complaint and escalation rate
- **AI quality guardrails:** hallucination/incorrect-answer rate from a sampled QA review, harmful or non-compliant response rate, repeated-question rate (proxy for failed understanding)

8. Analysis Plan

Success is defined as a statistically significant reduction in churn in treatment, supported by improved CSAT and meaningful chatbot adoption.

1. Compare churn between treatment and control using ITT; report effect size, 95% CI, and p-value
2. Evaluate CSAT differences at 95% confidence
3. Measure chatbot adoption, containment, and escalation rates within treatment
4. Run the triggered analysis on support-engaged users for CSAT and churn
5. Examine channel distribution (chat vs. call) to understand preference and substitution effects
6. Sample-based QA review of chatbot transcripts for accuracy, tone, and policy compliance
7. Segment results by tenure, prior support history, and issue type to surface heterogeneous effects

9. Decision Framework

Strong positive result Statistically significant churn reduction in treatment, supported by improved CSAT and meaningful chatbot adoption with acceptable AI quality guardrails. → Proceed with broader rollout and invest in feature and model optimization.

Mixed / no significant result No significant churn change, but directional improvements in CSAT, adoption, or containment. → Investigate barriers (awareness, AI accuracy, escalation friction, issue-type fit). Iterate on MVP gaps before re-testing rather than greenlighting rollout.

Negative result No improvement or regression in churn and/or CSAT, or guardrail breaches (e.g., elevated hallucination or complaint rate). → Do not proceed with rollout. Reassess whether the root cause is service quality, AI capability, or upstream product issues.

10. Risks & Limitations

- **Feature awareness:** Low visibility of the chatbot may suppress adoption and understate impact. Mitigation: in-app entry points and contextual prompts during support-seeking flows.
- **MVP limitations:** A simplified chatbot may underperform a production-grade AI, biasing CSAT and containment downward. Frame results as a lower bound on potential.
- **AI accuracy and hallucination risk:** Incorrect answers can damage trust and inflate churn. Mitigation: scoped intents for the MVP, conservative confidence thresholds, fast escalation to humans, transcript QA sampling.
- **Operational readiness:** Inadequate agent training on chat-to-human handoff may produce inconsistent service. Mitigation: pre-launch training and a dedicated escalation queue.
- **Novelty effect:** Early adopters may overuse or underuse the chatbot relative to steady state. Mitigation: minimum 6-week duration; report week-over-week trends.
- **External factors:** Churn may be driven by product or pricing issues unrelated to support. Mitigation: monitor external campaigns and outages; pre-register exclusion criteria for major incidents.
- **Data privacy and compliance:** GDPR, CCPA, and AI-specific disclosure requirements may affect logging, training data use, and user consent flows. Legal review required before launch.
- **Selection into channel:** Users who choose chat may differ systematically from call users. ITT is the primary metric for this reason; triggered analysis is interpretive only.

11. Out of Scope

- Web and desktop chatbot surfaces
 - Multilingual support beyond launch languages
 - Voice-based AI agent
 - Proactive (outbound) AI conversations
 - Full integration with downstream CRM workflows beyond what the MVP requires
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